

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
3	(a)			The smallest scale division on ruler is $[\pm]1$ cm (1) [so correct]			1	1		1
	(b)	(i)		d_1 scale 5 cm/division and d_2 scale 2 cm per division (1) All 5 points plotted correctly $\pm < 1$ small square tolerance (1) 4 or less points plotted correctly $\pm < 1$ small square tolerance (0) [Best fit] straight line between 5 – 40 cm for d_1 (1)		3		3	3	3
		(ii)		Gradient = $\frac{16}{40}$ ((1) selection of pair of values) alternatives accepted = 0.4 (1) $\frac{W_1}{W_2} = \frac{4}{10}$ (1) [so Mary is correct] Full marks can be awarded for this part if axes are plotted the wrong way round. Answer only of 0.4 award 1 mark only			3	3	3	3
	(c)	(i)		Substitution: 10×32 (1) = 320 [N cm] (1)	1	1		2	2	2
		(ii)		Anticlockwise moment = $320 = 4 \times d_1$ $d_1 = \frac{320}{4} = 80$ [cm] (1) Ruler not long enough or 80 cm > distance from pivot to ruler end or greater than 50 cm [so can't be balanced] (1) Alternative: maximum anticlockwise moment = $4 \times 50 = 200$ [N cm] (1) clockwise moment 320 N cm > maximum anticlockwise moment 200 N cm or 120 less than needed [so can't be balanced] (1)		2		2	2	2
				Question 3 total	1	6	4	11	10	11